Pawar Aravind

co23btech11016@iith.ac.in | pawaravipra04@gmail.com | +91 9392359190 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.66
RESONANCE JUNIOR COLLEGE, WARANGAL	Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION)	2023	96.10%
EKASHILA HIGH SCHOOL, WARANGAL	Class X (CENTRAL BOARD OF SECONDARY EDUCATION)	2021	95.80%

EXPERIENCE

Faculty-Guided Research Project - Parallel Multi-Objective Optimization for Smart Contracts

- **Developed** NSGA-II for blockchain transaction selection from **mempool**, optimizing profit, throughput, and block size.
- **Implemented** parallel multi-objective transaction selection, improving **scalability** and **computational efficiency.**
- **Explored blockchain transaction** and block generation, applying multi-objective optimization to block construction.

SKILLS

- **Programming Languages:** C++, C, Python, JavaScript, SQL
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Multithreading
- **Data Science / ML:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook
- **Web / Backend:** HTML, CSS, JavaScript, React.js, Node.js, Express.js, REST APIs
- **Databases:** MongoDB, PostgreSQL, MySQL
- **Tools / Technologies:** Git, GitHub, Linux, CMake, GoogleTest, Google Colab, VS Code, Postman

PROJECTS

IPL Auction Intelligence

Python | *Pandas* | *NumPy* | *Matplotlib* | *Gemini API* | *Groq API* | *Jupyter Notebook*

- **Built** an IPL auction analytics system using Python and Pandas for EDA, spending analysis, player rankings, and insights.
- **Engineered** a **natural-language query engine** using Gemini API, Pydantic, and validation for analytical queries.
- **Implemented** Pandas queries and Matplotlib visualizations with **Gemini->Groq fallback** and error handling.

InterviewGPT - AI-Powered Mock Interview Platform

React.js | *Node.js* | *Express.js* | *MongoDB* | *Gemini API* | *Groq API* | *Ollama* | *JWT* | *FFmpeg* | *REST APIs*

- **Developed adaptive AI interviews** from resumes and job descriptions using LLMs, enabling role-specific assessment.
- **Engineered voice interviews** using MediaRecorder, FFmpeg, STT, and TTS, enabling **automated skill evaluation**.
- **Architected Gemini->Groq->Ollama fallback** with an adaptive state machine, JWT, and MongoDB, enabling resilient context-aware interviews and personalized insights.
- **Designed** automated interview evaluation to assess technical correctness, problem solving, and communication skills.

MiniSearch -- Full-Stack Information Retrieval Search Engine

C++17 | React.js | REST API | Data Structures & Algorithms | TF-IDF | Information Retrieval | API Integration

- **Built** a C++ search engine with **text processing, inverted indexing, and Boolean&phrase query retrieval.**
- **Implemented** TF-IDF ranking to score and order documents, improving relevance for multi-document queries.
- **Developed** REST API and React.js frontend with Wikipedia fallback, result highlighting, & performance benchmarking.

Mini Linux Shell

C++17 | Linux | POSIX | Systems Programming | System Calls | Process Management | File Descriptors | CMake

- **Developed** a Linux shell in C++17 supporting **filesystem and process commands** via **POSIX system calls.**
- **Implemented** fork()-execvp()-waitpid() for process creation, execution, synchronization, and zombie prevention.
- **Built** filesystem and process tools using POSIX I/O, /proc inspection, signals, and system-call error handling.

ACHIEVEMENTS

- **Solved 600+ DSA problems** on GeeksforGeeks, covering algorithms, data structures, problem solving, and competitive programming; secured **Institute Rank under 40** and **World Rank under 15K**.
- **Maintained a 150+ day POTD streak** on GeeksforGeeks, demonstrating consistency and discipline.
- **Selected as Libero** for IIT Hyderabad's Volleyball Team for the upcoming Inter-IIT Sports Meet 2026 (IISM'26).

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator** -- Office of Career Services (OCS), IIT Hyderabad
- **Internship Coordinator** -- Office of Career Services (OCS), IIT Hyderabad
- **Hospitality Coordinator** -- Milan Inter-Hostel Competition, IIT Hyderabad

EXTRACURRICULAR & ADDITIONAL INFORMATION

- **NSO Volleyball Player:** Participated in IIT Hyderabad competitions including Milan, Diesta, and IVL.
- **Half Marathon Volunteer (2024):** Supported event coordination and logistics at IIT Hyderabad.
- **Hobbies & Interests:** Volleyball (Libero) | Athletics (100m-400m) | Cricket & IPL | Chess & Carrom | Anime & Movies
- **Relevant Courses:** Introduction to Programming, Data Structures and Applications, Operating Systems, Foundation of Machine Learning, Fundamentals of Scientific Computing, Computer-Oriented Numerical Methods, Calculus I & II, Transform Techniques, Differential Equations, Elementary Linear Algebra, Probability, Statistics, Complex Variables
- **Soft Skills:** Problem Solving, Analytical Thinking, Teamwork, Communication, Leadership, Adaptability.
- **Spoken Languages:** English, Hindi, Telugu.